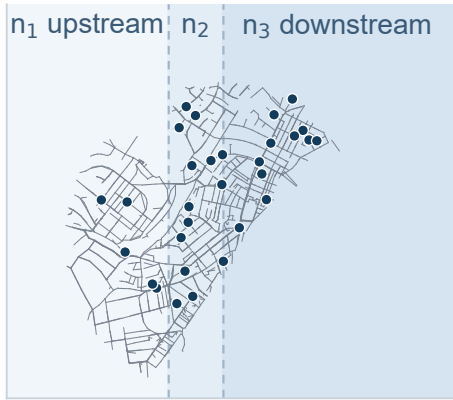
Pre-registered identifiability audit



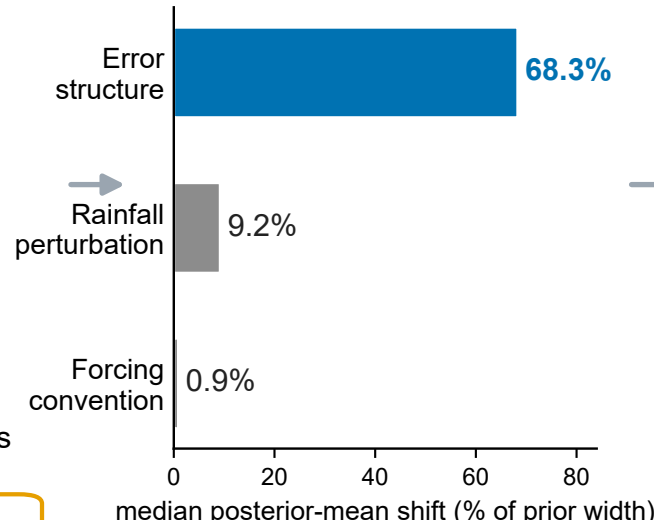
real drainage network
30 level gauges · 11 storm events

$\theta = (n_1, n_2, n_3)$ three roughness zones

E0 independent | E1 AR(1) | E2 AR(1)+bias

same data · same surrogate · three likelihoods

Error structure dominates



but grading ≠ prediction skill

gradings flipping across structures **6/33**

holdout tasks passing the 5% CRPS bar **0/33**

leave-one-event-out prediction, pre-registered threshold

Grading gains and posterior contraction do not, by themselves, demonstrate prediction quality.